

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Light - Reflection and Refraction

CBSE Science 2026-27 | Board-examinable

Exactly 20 numericals followed by 10 theory questions, with step-based answers for Class X Physics revision.

Question 1

An object is 30 cm in front of a concave mirror of focal length 15 cm. Find image distance and magnification.

Answer:

Using $u = -30$ cm and $f = -15$ cm: $1/f = 1/v + 1/u$ gives $v = -30$ cm. $m = -v/u = -1$. The image is real, inverted and same-sized.

Question 2

An object is 60 cm in front of a concave mirror of focal length 20 cm. Find image distance.

Answer:

$u = -60$ cm, $f = -20$ cm. $1/v = -1/20 + 1/60 = -1/30$, so $v = -30$ cm.

Question 3

An object is 15 cm in front of a concave mirror of focal length 10 cm. Find image distance.

Answer:

$u = -15$ cm, $f = -10$ cm. $1/v = -1/10 + 1/15 = -1/30$, so $v = -30$ cm.

Question 4

A concave mirror forms a real image 24 cm from it when the object is 40 cm away. Find focal length.

Answer:

$u = -40$ cm, $v = -24$ cm. $1/f = 1/v + 1/u = -1/24 - 1/40 = -1/15$, so $f = -15$ cm.

Question 5

A convex mirror has focal length 15 cm. An object is 30 cm in front. Find image distance.

Answer:

$u = -30$ cm, $f = +15$ cm. $1/v = 1/15 + 1/30 = 1/10$, so $v = +10$ cm.

Question 6

A concave mirror forms an image 3 times the object size. If image height is -12 cm, find object height.

Answer:

$m = h_i/h_o = -3$. Thus $-12/h_o = -3$, so $h_o = 4$ cm.

Question 7

A convex lens of focal length 20 cm has an object at 60 cm. Find image distance.

Answer:

$u = -60$ cm, $f = +20$ cm. $1/v = 1/20 - 1/60 = 1/30$, so $v = +30$ cm.

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Question 8

An object is 30 cm from a convex lens of focal length 10 cm. Find image distance.

Answer:

$u = -30$ cm, $f = +10$ cm. $1/v = 1/10 - 1/30 = 1/15$, so $v = +15$ cm.

Question 9

A concave lens has focal length 15 cm. An object is 30 cm away. Find image distance.

Answer:

$u = -30$ cm, $f = -15$ cm. $1/v = -1/15 - 1/30 = -1/10$, so $v = -10$ cm.

Question 10

A convex lens forms a real image 24 cm away when the object is 40 cm away. Find focal length.

Answer:

$u = -40$ cm, $v = +24$ cm. $1/f = 1/24 - 1/(-40) = 1/15$, so $f = +15$ cm.

Question 11

A lens gives magnification -2.5 for an object 20 cm away. Find image distance.

Answer:

$m = v/u = -2.5$. Therefore $v = (-2.5)(-20) = +50$ cm.

Question 12

A lens has power +2.5 D. Find focal length.

Answer:

$P = 1/f$ (in metres). $f = 1/2.5 = 0.4$ m = 40 cm.

Question 13

Find power of a concave lens of focal length 50 cm.

Answer:

$f = -0.50$ m. $P = 1/f = -2$ D.

Question 14

For a glass medium, $i = 45$ degrees and $r = 30$ degrees. Find refractive index using $n = \sin i / \sin r$.

Answer:

$n = \sin 45 / \sin 30 = (\sqrt{2}/2) / (1/2) = \sqrt{2}$, about 1.41.

Question 15

Speed of light in glass is 2.0×10^8 m/s. Find refractive index of glass.

Answer:

$n = c/v = (3.0 \times 10^8) / (2.0 \times 10^8) = 1.5$.

Question 16

A lens has power -4 D. Find focal length.

Answer:

$f = 1/P = 1/(-4) = -0.25$ m = -25 cm.

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Question 17

A convex lens of focal length 25 cm forms an image at 50 cm. Find object distance.

Answer:

$1/u = 1/v - 1/f = 1/50 - 1/25 = -1/50$. Thus $u = -50$ cm.

Question 18

A concave mirror of focal length 12 cm forms an image at 36 cm. Find object distance.

Answer:

$1/u = 1/f - 1/v = -1/12 + 1/36 = -1/18$. Thus $u = -18$ cm.

Question 19

An object 4 cm high forms a real image 12 cm high in a lens. Find magnification.

Answer:

$m = h_i/h_o = -12/4 = -3$. The negative sign shows that the image is inverted.

Question 20

A ray enters glass at 30 degrees and refracts at 19.5 degrees. Find n approximately.

Answer:

$n = \sin 30 / \sin 19.5$ about $0.5/0.334 = 1.50$.

Question 21

State the two laws of reflection.

Answer:

The incident ray, reflected ray and normal lie in one plane. The angle of incidence equals the angle of reflection.

Question 22

State the two laws of refraction.

Answer:

The incident ray, refracted ray and normal lie in one plane. For a pair of media, $\sin i / \sin r$ is constant.

Question 23

Why does light bend towards the normal on entering glass from air?

Answer:

Glass is optically denser, so light slows down and bends towards the normal.

Question 24

Differentiate a real and a virtual image.

Answer:

Real rays actually meet and the image can be obtained on a screen. Virtual rays appear to meet and cannot be obtained on a screen.

Question 25

What is the principal focus of a concave mirror?

Answer:

It is the point on the principal axis where rays parallel to the axis meet after reflection.

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Question 26

Why is a convex mirror used as a rear-view mirror?

Answer:

It gives an erect, diminished image and a wide field of view.

Question 27

How does a convex lens behave for an object within its focal length?

Answer:

It forms a virtual, erect and magnified image on the same side as the object.

Question 28

What is magnification for a mirror and lens?

Answer:

For a mirror $m = -v/u = h_i/h_o$. For a lens $m = v/u = h_i/h_o$.

Question 29

Why does a prism disperse white light?

Answer:

Different colours refract by different amounts in glass, separating white light into its spectrum.

Question 30

How is power of a lens related to focal length?

Answer:

Power $P = 1/f$ when f is in metres. A convex lens has positive power and a concave lens has negative power.

Sources: CBSE Science curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.